Garrett Goodwin

Software Engineer | Unity Developer | Gameplay Programmer

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PROFESSIONAL SUMMARY

Software engineer and Unity developer with experience building real-time desktop, mixed-reality, and game applications. Combines professional aircraft-data visualization work at GTRI with end-to-end ownership of a substantial independent Unity title. Strong in C#, C++, data-driven architecture, performance optimization, debugging, and developer tooling.

TECHNICAL SKILLS

Languages: C#, C++, Python, Java

Unity and Platforms: Unity 2D/3D, URP, ScriptableObjects, New Input System, Cinemachine, Photon PUN 2, Microsoft HoloLens

Engineering: Object-oriented design, data-driven architecture, debugging, performance optimization, automated benchmarking, Git, Agile/Scrum

Tools: GitHub, Visual Studio, Blender, Photoshop, Trello

EXPERIENCE

Independent Game Developer | ReapKeep *Jan 2025 - Present*

Solo project | Unity, C#, 2D URP | Planned for Steam

- Independently design and develop a 2D reverse bullet-hell roguelike, owning software architecture, gameplay systems, UI, technical tools, balancing, and production planning.
- Architected modular, data-driven systems for four characters, weapons and evolutions, items, enemies, three difficulty profiles, progression, saves, achievements, and UI using C# and ScriptableObjects.
- Built repeatable gameplay benchmarks that export timestamped CSV telemetry for frame rate, enemy population, damage, economy, progression, encounter timing, and survival.
- Created reusable Unity editor tooling for CSV import/export, asset generation, stable-ID migrations, content audits, diagnostics, and Play Mode debugging.
- Optimized high-volume combat with object pooling, non-allocating physics queries, UI prewarming, rate-limited effects, bounded populations, and lifecycle-safe cleanup.

Research Intern | Georgia Tech Research Institute *Jun 2023 - Dec 2024*

Arizona, United States | Unity and Microsoft HoloLens

- Developed Unity features that replayed recorded aircraft data as interactive visualizations on desktop and Microsoft HoloLens.
- Investigated and resolved cross-platform rendering, performance, and usability issues while adapting features to mixed-reality device constraints.
- Collaborated with team members through iterative reviews, testing, and troubleshooting to improve visualization accuracy and application stability.

SELECTED PROJECTS

Upturn | Unity, C#, Photon PUN 2 *2022 - 2023*

- Built a networked 3D first-person shooter with multiplayer synchronization and modular weapon systems.

Light in the Dark | Unity, C# *2021 - 2022*

- Created a 2D wave shooter with custom enemy AI, combat mechanics, and a multi-stage weapon upgrade system.

EDUCATION

Arizona State University *Dec 2024*

Bachelor of Science in Computer Science | GPA: 3.78